# BME5006 Advanced Topics In Biomedical Engineering – Homework (Lab Test):

# Effect of Smoking Status on Lung Cancer and Myocardial Infarction

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## 1- Introduction

This study examined whether smoking was associated with two major health outcomes: lung cancer and myocardial infarction (MI). The analysis used a dataset of 500 subjects. The predictors were age, smoking history, and smoking duration. The outcomes were lung cancer and MI. The main aim was to describe the sample, explore possible patterns, test whether the observed associations were statistically significant, and estimate the strength of these associations with logistic regression.

## 2- Methods

The dataset contained 500 observations and 6 variables. There were no missing values in any column. A small data correction was required because two entries in the lung cancer variable were recorded as text rather than as numeric codes. After this correction, the categorical variables were converted to factors in R. All the outputs and plots are obtained by RStudio program.

Descriptive statistics were calculated for age and for the categorical variables. Cross-tabulations were used to compare smoking status and smoking duration with MI and lung cancer. Exploratory graphs were created to support the descriptive findings.

For hypothesis testing, chi-square tests were used when expected cell counts were acceptable. Fisher's Exact Test was used when cell counts were less than 5. Logistic regression models were then fitted to estimate odds ratios (ORs) and 95% confidence intervals (CIs). Both unadjusted models and age-adjusted models were examined. Moreover, age variable is scaled in 10-year units so each age OR represents the change in odds per 10-year increase, for an additional analysis.

## 3- Results

### 3.1 - Data Quality and Preparation

The dataset included 500 patients and no missing values were detected. This means the full sample could be used in all analyses. One coding issue in the lung cancer variable was corrected before the statistical analysis began.

### 3.2 - Descriptive Statistics

The mean age of the sample was 55.3 years, with a median of 55 years, a standard deviation of 9.73 years, and a range from 30 to 87 years. These values show that the sample was mainly middle-aged to older adults. Shown in Figure 1.

The smoking distribution showed that 210 patients (42.0%) had never smoked and 290 patients (58.0%) had smoked at some point. For smoking duration, 210 patients were in the never-smoking group, 63 were in the less than 1 year group, 80 were in the 2 to 5 years group, and 147 were in the more than 5 years group. Figures 2 and 3.

As shown in Figure 4 and 5, the outcome distribution showed that 140 patients had MI and 76 patients had lung cancer. This means MI was more common than lung cancer in this dataset. In other words, number of lung cancer cases is less than MI cases.

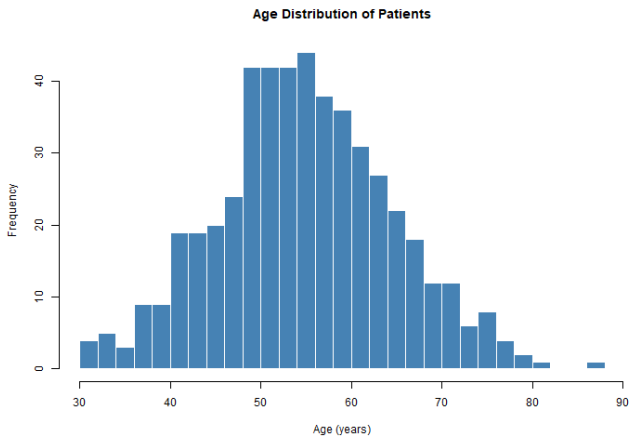


Figure 1: Age distribution of the patients.

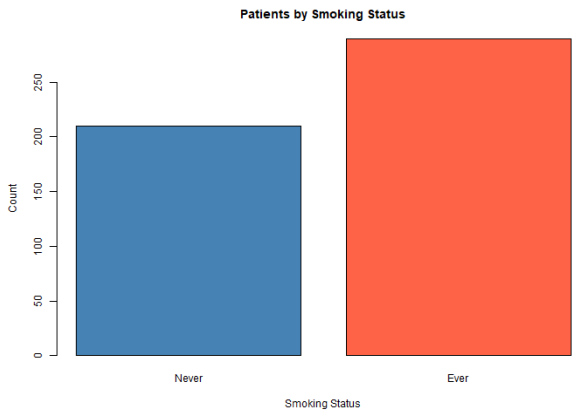

Figure 2: Number of patients by smoking status.

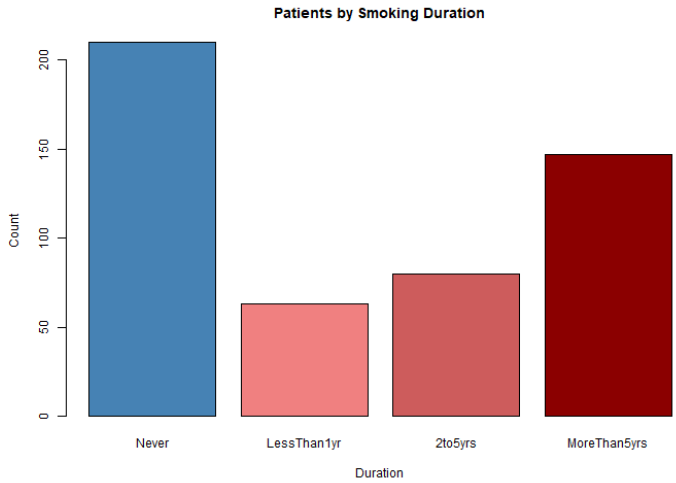


Figure 3: Number of patients by smoking duration.

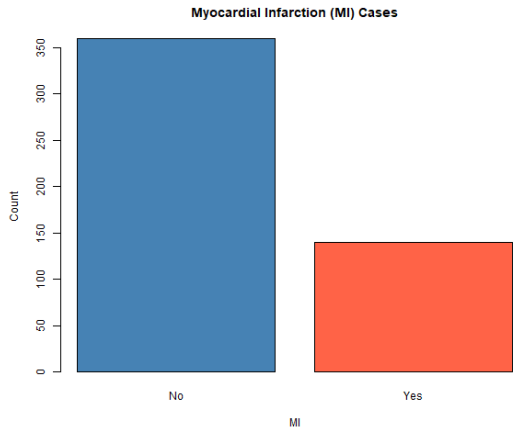


Figure 4: Number of myocardial infarction cases.

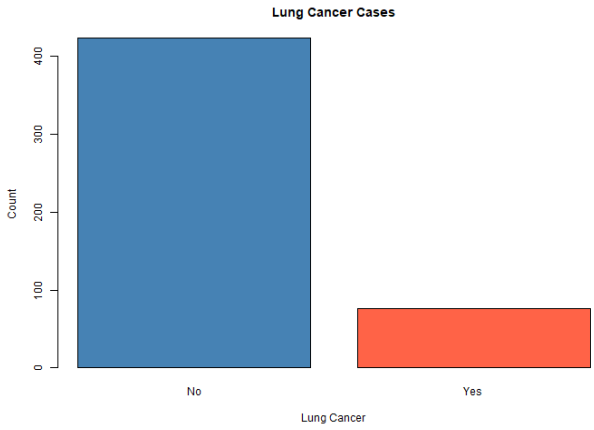


Figure 5: Number of lung cancer cases.

### 3.3 - Exploratory Analysis of Associations

The cross-tabulations (plotted as bar graphs with percentages here) showed clear differences between smoking groups. Among never-smokers, 27 of 210 patients had MI, which corresponds to 12.9%. Among ever-smokers, 113 of 290 patients had MI, or 39.0%. This suggests that MI was much more common in patients with a history of smoking. Shown in Figure 6.

The difference for lung cancer was even stronger. Among never-smokers, only 4 of 210 patients had lung cancer, which is 1.9%. Among ever-smokers, 72 of 290 patients had lung cancer, or 24.8%. This shows a large increase in lung cancer frequency among patients who had smoked. Shown in Figure 7.

Figure 8 shows that MI was present in 12.9% of the never-smoking group, 44.4% of the less than 1 year group, 36.2% of the 2 to 5 years group, and 38.1% of the more than 5 years group. Lung cancer was present in 1.9% of the never-smoking group, 25.4% of the less than 1 year group, 21.2% of the 2 to 5 years group, and 26.5% of the more than 5 years group, as seen in Figure 9. These results show that all smoking-duration groups had higher outcome rates than the never-smoking group. However, the percentages across smoking-duration categories were not perfectly stepwise.

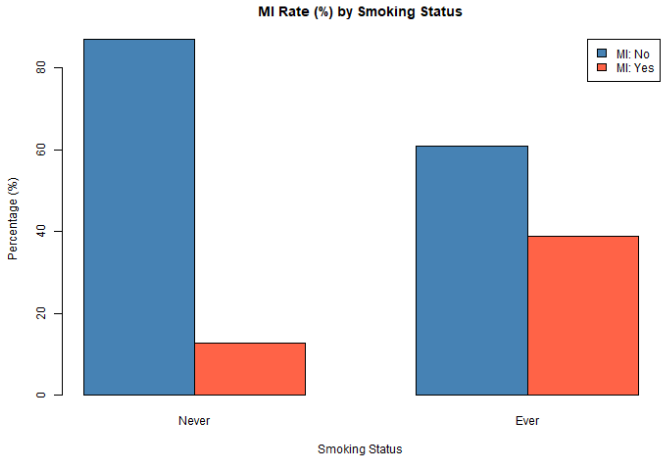


Figure 6: Percentage of MI cases by smoking status.

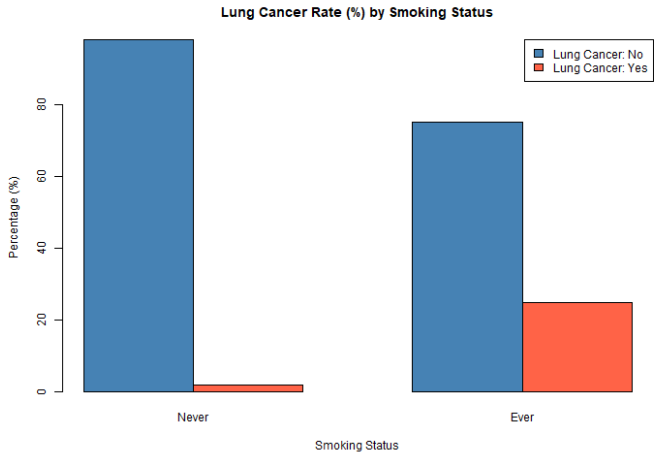


Figure 7: Percentage of lung cancer cases by smoking status.

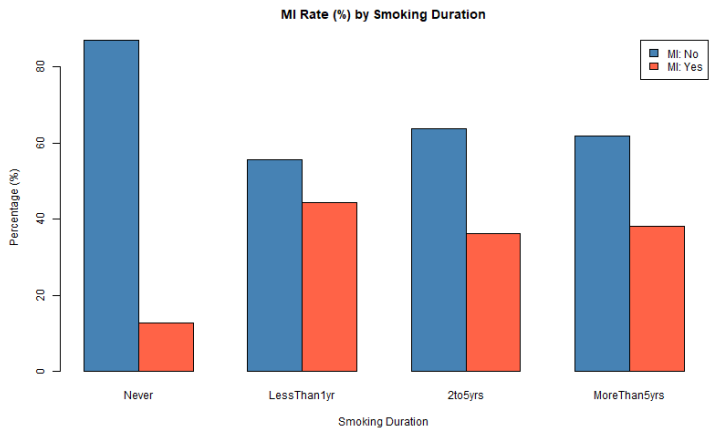


Figure 8: Percentage of MI cases by smoking duration.

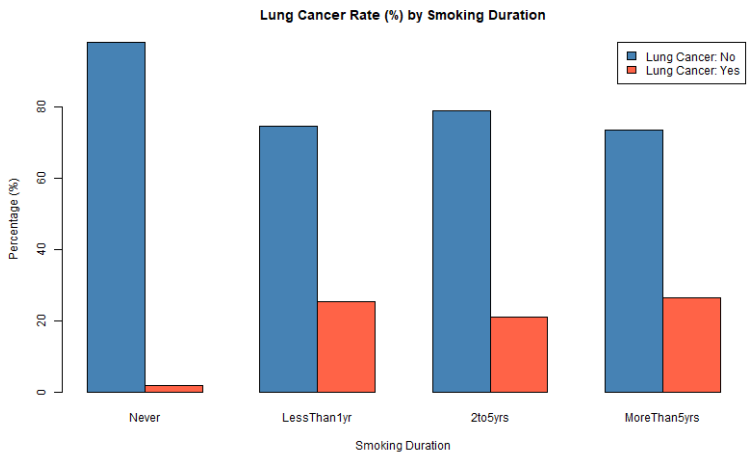


Figure 9: Percentage of lung cancer cases by smoking duration.

## 4- Hypothesis Testing

The null hypothesis for each comparison was that there was no association between smoking and the outcome. The alternative hypothesis was that an association was present. Chi-square was used for the MI comparisons. Fisher's Exact Test was used for the lung cancer comparisons because the observed tables included counts below 5.

Table 1. Hypothesis testing results.

| Comparison | Test | Test statistic | df | p-value |
|---|---|---|---|---|
| Smoking status vs MI | Chi-square | 39.898 | 1 | <0.001 |
| Smoking status vs lung cancer | Fisher's Exact Test | OR = 16.951<br>95% CI = 6.170 to 65.015 | Not applicable | <0.001 |

| | | | | |
|---|---|---|---|---|
| Smoking duration vs MI | Chi-square | 42.469 | 3 | <0.001 |
| Smoking duration vs lung cancer | Fisher's Exact Test | Absent in R software | Not applicable | <0.001 |

All four comparisons were statistically significant. These results indicate that MI and lung cancer frequencies differed by smoking status and by smoking duration in this dataset.

## 5 – Logistic Regression

### 5.1 Unadjusted Models

Odds ratios, 95% confidence intervals, and p-values were estimated from logistic regression models. In the unordered smoking-duration models, each duration group was compared with the never-smoking group. In the ordered models, the linear trend term tested whether the outcome changed across the ranked duration categories.

Table 2. Unadjusted logistic regression results.

| Model | Term | OR | 95% CI | p-value |
|---|---|---|---|---|
| Model 1: Smoking status -> MI | Smoking Status: Ever vs Never | 4.327 | 2.746 to 7.021 | <0.001 |
| Model 2: Smoking status -> Lung Cancer | Smoking Status: Ever vs Never | 17.009 | 6.899 to 56.565 | <0.001 |
| Model 3a: Smoking duration -> MI | Smoking Duration: LessThan1yr vs Never | 5.422 | 2.867 to 10.364 | <0.001 |
| Model 3a: Smoking duration -> MI | Smoking Duration: 2to5yrs vs Never | 3.854 | 2.099 to 7.127 | <0.001 |
| Model 3a: Smoking duration -> MI | Smoking Duration: MoreThan5yrs vs Never | 4.171 | 2.493 to 7.124 | <0.001 |
| Model 3b: Smoking duration -> MI | Smoking Duration: Linear trend (.L) | 2.415 | 1.656 to 3.563 | <0.001 |
| Model 4a: Smoking duration -> Lung Cancer | Smoking Duration: LessThan1yr vs Never | 17.532 | 6.110 to 63.376 | <0.001 |

| | | | | |
|---|---|---|---|---|
| Model 4a: Smoking duration -> Lung Cancer | Smoking Duration: 2to5yrs vs Never | 13.897 | 4.935 to 49.652 | <0.001 |
| Model 4a: Smoking duration -> Lung Cancer | Smoking Duration: MoreThan5yrs vs Never | 18.597 | 7.250 to 63.204 | <0.001 |
| Model 4b: Smoking duration -> Lung Cancer | Smoking Duration: Linear trend (.L) | 6.745 | 3.493 to 15.570 | <0.001 |
| Model 5: Age -> MI | Age: per 10-year increase | 2.820 | 2.204 to 3.670 | <0.001 |
| Model 6: Age -> Lung Cancer | Age: per 10-year increase | 1.834 | 1.413 to 2.405 | <0.001 |

In the unadjusted models, smoking status and smoking duration were associated with both outcomes. Age was also associated with both MI and lung cancer before adjustment.

**5.2 Age-Adjusted Models**

Table 3. Age-adjusted logistic regression results.

| Model | Term | Adjusted OR | 95% CI | p-value |
|---|---|---|---|---|
| Model 7: Smoking status + age -> MI | Smoking Status: Ever vs Never | 2.314 | 1.401 to 3.899 | 0.001 |
| Model 7: Smoking status + age -> MI | Age: per 10-year increase | 2.410 | 1.854 to 3.180 | <0.001 |
| Model 8: Smoking status + age -> Lung Cancer | Smoking Status: Ever vs Never | 13.768 | 5.402 to 46.684 | <0.001 |
| Model 8: Smoking status + age -> Lung Cancer | Age: per 10-year increase | 1.290 | 0.965 to 1.732 | 0.087 |
| Model 9a: Smoking duration + age -> MI | Smoking Duration: LessThan1yr vs Never | 2.866 | 1.435 to 5.736 | 0.003 |
| Model 9a: Smoking duration + age -> MI | Smoking Duration: 2to5yrs vs Never | 2.299 | 1.196 to 4.419 | 0.012 |

| | | | | |
|---|---|---|---|---|
| Model 9a: Smoking duration + age -> MI | Smoking Duration: MoreThan5yrs vs Never | 2.110 | 1.194 to 3.769 | 0.011 |
| Model 9a: Smoking duration + age -> MI | Age: per 10-year increase | 2.414 | 1.855 to 3.189 | <0.001 |
| Model 9b: Smoking duration + age -> MI | Smoking Duration: Linear trend (.L) | 1.571 | 1.039 to 2.387 | 0.033 |
| Model 9b: Smoking duration + age -> MI | Age: per 10-year increase | 2.414 | 1.855 to 3.189 | <0.001 |
| Model 10a: Smoking duration + age -> Lung Cancer | Smoking Duration: LessThan1yr vs Never | 14.156 | 4.778 to 52.282 | <0.001 |
| Model 10a: Smoking duration + age -> Lung Cancer | Smoking Duration: 2to5yrs vs Never | 11.699 | 4.063 to 42.401 | <0.001 |
| Model 10a: Smoking duration + age -> Lung Cancer | Smoking Duration: MoreThan5yrs vs Never | 14.966 | 5.628 to 51.999 | <0.001 |
| Model 10a: Smoking duration + age -> Lung Cancer | Age: per 10-year increase | 1.278 | 0.954 to 1.719 | 0.101 |
| Model 10b: Smoking duration + age -> Lung Cancer | Smoking Duration: Linear trend (.L) | 5.885 | 2.985 to 13.763 | <0.001 |
| Model 10b: Smoking duration + age -> Lung Cancer | Age: per 10-year increase | 1.278 | 0.954 to 1.719 | 0.101 |

After adjustment for age, smoking status remained associated with MI and lung cancer. Age remained associated with MI in the adjusted models. In the lung cancer models that included smoking status or smoking duration, the age coefficient was not statistically significant.

## 6- Discussion

The hypothesis tests and logistic regression models were consistent. Smoking status was associated with both MI and lung cancer. Smoking duration was also associated with both outcomes in the unordered and ordered models.

The odds ratios were larger for lung cancer than for MI. In the age-adjusted models, the association between smoking and lung cancer remained large. For MI, the smoking coefficients were smaller after age adjustment, while the age coefficient remained statistically significant.

The duration models can be read in two ways. The unordered models compare each duration category with the never-smoking group. The ordered models test a linear trend across the ranked duration categories. Both approaches supported an association in this dataset.

This analysis is based on observational data. The results show association, not causation. Other variables not included in the dataset may also affect MI and lung cancer.

## 7- Conclusion

In this sample of 500 patients, smoking status and smoking duration were associated with both MI and lung cancer. The associations with lung cancer were larger than the associations with MI. Age was also associated with MI and remained statistically significant after adjustment.

**Short Public Summary**

This study used data from 500 patients to examine smoking, heart attack, and lung cancer. The results showed that smoking status was associated with both myocardial infarction and lung cancer. Smoking duration was also associated with both outcomes. The associations were larger for lung cancer than for myocardial infarction. Age was related to myocardial infarction and remained important after adjustment. In the lung cancer models, age was less important after smoking was included. These findings show a clear relationship between smoking and the two health outcomes in this dataset. However, the data are observational, so the results should be interpreted as associations rather than proof of causation.